Historical Computing: Evolution of Human-Computer Interaction

In Partial Fulfillment of the
Course Requirements for

INTRODUCTION TO COMPUTER ORGANIZATION AND ARCHITECTURE 2 (CSARCH2)
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Submitted by Group 8 [S02]:
Helaga, Raina
Lee, Jason Benedict
Maravilla, Sofia
Plurad, Venice Raeka
Ramirez, Diana Angela

Submitted to:
Roger Luis Uy

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GITHUB REPOSITORY LINK

Link to the group's repository: https://github.com/ventilogz/-CSARCH2-Case-Study-Project-2

I. VIRTUAL EXHIBIT PLAN

This proposal presents the planned virtual exhibit on Historical Computing, specifically focusing on the evolution of Human-Computer Interaction (HCI). It outlines the exhibit's content, structure, interactive elements, technical implementation, and design approach.

A. Introduction [Hardware/software deep-dives]

Human-Computer Interaction (HCI) delves into how people interact and use computers and technology. Over the years, the way humans use computers has evolved significantly, from relying on punch cards and command-line inputs to advanced technologies like voice assistants. These developments have made computers more accessible, intuitive, and efficient for users. Understanding the evolution of HCI helps us appreciate how technology has made computers more accessible, intuitive, and user-friendly. It also shows how past innovations continue to shape the future of computing.

To further explore this topic, this proposal will discuss the key milestones and developments that shaped the evolution of HCI, highlighting how each stage transformed the way humans interact with computers. Specifically the exhibit will cover the following stages:

1. Batch Processing & Punch cards
2. Command Line Interfaces (CLI)
3. Pointing Devices
4. The Graphical User Interface (GUI) era
5. Touch and Mobile Interfaces
6. Spatial & Voice Computing

Through Interactive elements and visual materials, this exhibit aims to give users a better understanding of the key developments in HCI and how they changed the way people interact with computers throughout history.

B. Main Exhibit

1. **1940s-1950s: Batch Processing & Punch Cards (ENIAC, physical rewiring).**
    a. **Overview:** The very first in the development of computers was batch processors and punchcards. During the 1940s and 1950s, computers used punch cards as the main method of input. Users would create instructions by punching holes into cards which will be processed in batches. Programming was also difficult since making changes often required physically rewiring the machine.
    b. **Key Topics:**
        - Punch Cards
        - Batch Processing
        - ENIAC
        - Physical rewiring

- Early programming

c. **HCI Significance:**

Batch processing and punch card were among the earliest forms. During this time, interaction between humans and computers was very limited since users could not communicate directly with the machine. Although the process was slow and time-consuming, it laid the foundation for future developments that made computers more interactive and easier to use.

2. **1960s-1970s: Command Line Interfaces (CLI) (Teletype, keyboards, early Unix).**

a. **Overview:**
   - Following the use of punch cards, Command-Line Interfaces (CLI) became the next major development in Human-Computer Interaction, enabling users to interact with computers by entering text-based commands through a keyboard. In the 1960s, minicomputers became more widely used and allowed multiple users to access a computer system. However, due to limited computing resources and the lack of graphical interfaces, users relied on Command-Line Interfaces (CLIs) to interact with computers through text-based commands.

b. **Key Topics:**
   - Text-based commands
   - Keyboards and Terminals
   - Early Unix systems
   - File Management

c. **HCI Significance:**
   - CLIs played an important role in the evolution of HCI because it allowed users to interact with computers directly through typed commands. Users did not need to wait for their inputs to be processed, making computer use faster and more efficient. This development paved the way for more advanced interfaces.

3. **1968: Pointing devices (Douglas Engelbart introduces the mouse).**

a. **Overview:**
   - The idea of the computer mouse was proposed by Douglas Engelbart and prototyped by Bill English at Stanford Research Institute. This technology revolutionized navigation and was officially patented. It was publicly debuted during the groundbreaking event demonstration that introduced many foundational elements of modern computing, “Mother of All Demos”.

b. **Key Topics:**

- Comparative input testing between light pen, joystick and knee operated devices to prove the mouse's superiority in aspects such as speed and accuracy.
- The progression from the initial wooden prototype to the three-button model.
- The concurrent development of a five-key "keyset," designed to be used with the left hand for entering text and commands while the right hand remained on the mouse.

c. **HCI Significance:**
- The computer mouse is a foundational milestone in Human-Computer Interaction. Developed as a core component of Douglas Engelbart's vision to "augment human intellect," it proved to be the era's most efficient pointing device. By bridging the gap between physical movement and digital execution, the mouse drove the transition from abstract, batch-processing computing to interactive display workstations, establishing the foundation for modern graphical user interfaces and collaborative digital environments.

4. **1980s: The GUI Era (Xerox PARC, Apple Macintosh, Windows).**

a. **Overview:**
- During the 1980s, GUI (Graphical User Interface) gained popularity, transforming the way computers were utilized and moving away from the command line interface to graphical and icon-driven interactions. That was the period when computers were becoming more intuitive to non-technical users with the introduction of the use of windows, menus, and pointers.

b. **Key Topics:**
- The introduction of windows, icons, menus (WIMP model).
- Using mouse control instead of typing commands
- Desktop metaphor (files, folders, trash bin)

c. **HCI Significance:**
- It was during this time that direct manipulation interfaces were established, which became the foundation of modern human-computer interaction. It helped to lessen cognitive burden, improve learnability, and introduce some of the principles of usability that are still applied to designing interfaces today.

5. **2000s: Touch & Mobile/Graphical Interfaces (Capacitive touchscreens, iPhone).**

a. **Overview:**
- The 2000s transformed Human-Computer Interaction (HCI) by shifting the focus from traditional desktop computing to mobile, multi-touch interfaces. Advancements in capacitive sensing replaced the single-point mouse cursor with direct physical manipulation, enabling both intuitive individual use and simultaneous multi-user collaboration. This era culminated in the launch of the iPhone, successfully commercializing

years of academic research and fundamentally redefining how society interacts with digital technology.

b. **Key Topics:**
   - Shift beyond single-touch resistive screens to capacitive screens that are capable of registering multiple simultaneous points of contact.
   - Optimization of the enabled touchscreen to become more accurate, durable and capable of swiping and pinching.
   - Development of large-scale touch surfaces that could distinguish between different users interacting at the exact same time.

c. **HCI Significance:**
   - This prompted the transition from indirect desktop computing to direct physical manipulation. By establishing a universal vocabulary of complex touch gestures and integrating sensors that adapted to physical movements, these interfaces drastically reduced user cognitive load.

6. **Present/Future: Spatial & Voice (AR/VR, Siri/Alexa, Neuralink/BCI).**
   a. **Overview:** In present day, and for future generations of technology to come, Spatial Computing and Voice User Interfaces is redefining how we interact with or how we command machines. This innovation moves us away from direct and physical manipulation and towards virtual or even conversational interfaces.
   b. **Key Topics:**
      - Merging physical and virtual spaces, from fully closed Virtual Reality (VR) to Mixed Reality (MR).
      - Spatial Mapping where sensors continuously scan your surroundings to detect and understand walls, objects, and obstacles.
      - Voice User Interfaces systems that swap visual assets for acoustic ones, relying on Automatic Speech Recognition to capture voice and Natural Language Processing to parse human meaning.
      - Conversational Delegation where we move past strict command strings to natural dialogue where an agent can process sentences based on tone, pronouns, implicit intent, etc.
   c. **HCI Significance:** Spatial Computing natural interactions through gestures, voice, commands, and spatial mapping. Users of Spatial Computing can manipulate 3D objects, collaborate in virtual spaces, and access information contextually, as if digital content is part of their reality. Voice User Interfaces shift the user experience from physical, visual, click or gestural-based interfaces to dynamic, conversational dialogues. It transforms computers from static tools into active digital translators, changing how people interact and command machines, process feedback, and form social relationships with technology

C. Conclusion

- The primary objective of this virtual exhibit is to grasp the rapid evolution of Human-Computer Interaction. By tracing the progression from early punch cards to modern spatial computing, the project demonstrates how technological advancements have progressively reduced cognitive load and made it accessible for public use. The future of HCI is rapidly advancing towards ambient computing. As spatial technologies, conversational AI, and Brain-Computer Interfaces mature, they create further research opportunities into how profoundly to integrate technology into the human experience.

By the end of this exhibit, readers will understand the evolution of Human-Computer Interaction from early punch cards to intuitive touchscreens. The project demonstrates how each milestone progressively reduced user cognitive load, eliminating the friction between human intent and machine execution to democratize computing. By establishing this historical context, readers are expected to recognize how the continuous drive for seamless interaction directly shapes the future of spatial computing, conversational AI, and anticipatory environments.

II. TECHNICAL STACK PLAN

1. **Base Framework and Runtime**
    a. Our group will adhere to the required templated specifications, using Astro 6 running on Node.js 26 for the base of our exhibit. Astro is a good framework for content-heavy sites and handles all our routing and generation. For the content itself, MDX (Markdown Extended) will be used. MDX allows us to write our main text using simple markdown while also letting us drop interactive components directly into the page.

2. **Code Logic**
    a. To power our interactive components, we will use React wrapped in TypeScript. TypeScript gives us strict type-safety, ensuring that the code for our interactive elements like stage management, user inputs, and logic is clean and bug-free.

3. **Styling**
    a. For the styling of the exhibit, our group plans to utilize Radix Primitives and Tailwind CSS. Radix Primitives is a low-level UI component library with a focus on accessibility, customization, and developer experience. Our group will use this as the base layer of our design to handle the foundational logic required for advanced UI elements. Since the components of Radix Primitives come with zero default styles, our group will use Tailwind CSS for layout, design, and theming. Tailwind CSS is a utility-first CSS framework used to rapidly style web applications, so instead of coding the general CSS stylesheets, we will style React elements directly.

4. **Animations**
    a. Animations will play a big role in our group's exhibit due to our planned interactive element which is an interactive timeline. For general icons, vector-based SVG graphics will be used for the exhibit from Lucide Icons, which is an open-source icon library, offering thousands of highly optimized SVG icons. For the animations of the interactive element and the exhibit itself, our group will use Motion (also known as Framer Motion)

which is an open-source animation and gesture library built for React. Motion's declarative API simplifies adding smooth, physics-based, and highly interactive UI animations which would work perfectly with our planned interactive elements

III. INTERACTIVE ELEMENT PLAN

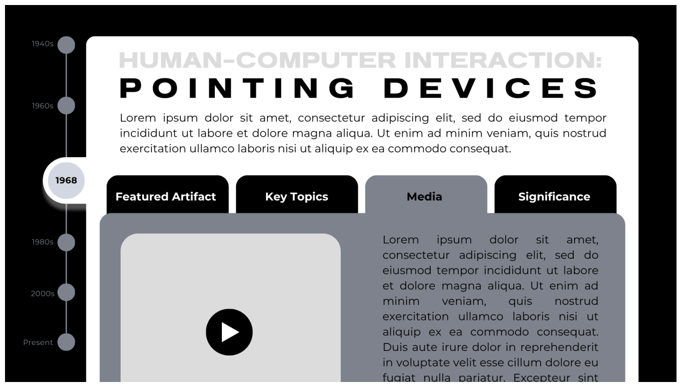
1. **Interactive Timeline** - The Interactive Timeline is a navigation tool that allows the user to traverse the development of HCI by clicking on any of the eras on a timeline. When you select each one, it will update the information in the content panel on the right, and will also provide information in a structured format, depending on the era.
    a. Features :
        i. Clickable milestones - Users click on an era (1940s–Present)..
        ii. Dynamic content switching - Content updates on the fly based on selected era, without reloading page.
        iii. Content sections are organized into tabs - Information is broken into Featured Artifact, Key Topics, Media, and Significance.
        iv. Active milestone highlighting - The selected era is highlighted so it is easy to navigate.
    b. User Flow :
        i. The vertical timeline shows eras, with the user selecting one.
        ii. The overview and content details for the selected era are updated in the right panel.
        iii. Users alternate between tabs to explore various aspects of the era.
2. **Simplified Punch Card Simulator** - A simple punch card simulator that shows how information was stored in early computers with punched cards. A short word or number is entered, and the corresponding holes are shown on a digital punch card, which is a visual representation of early data encoding techniques.
    a. Features :
        i. Short text input — A text field that allows users to type in a word or a number.
        ii. Digital punch card display — small punch card display grid (10-15 columns) that represents the encoded input.
        iii. Dynamic hole generation — Hole patterns are generated automatically depending on the characters that are entered.
        iv. Simplified encoding model — Representation of punch card data storage, without being concerned with a historical model.
    b. User Flow
        i. The user types a word or number of a limited size in the entry box.
        ii. The system transforms input to a simplified punch pattern.
        iii. The holes are shown on the digital punch card.
        iv. The user changes the input and sees that the punch pattern changes accordingly based upon the data.

3. **CLI Terminal Simulator** - A command-line simulation of an early computing interface that allows for user interaction via text rather than graphical commands. The system reacts to preset responses, similar to the way that early computers were used prior to the invention of the GUI.
    a. Features:
        i. Command Based Input - Users type simple commands.
        ii. Pre-scripted responses - Outputs are predetermined and given in response to certain inputs
        iii. Simple interface - Input box and console-like output display is used.
        iv. Conditional logic system - Constructed with simple if/else command handling.
    b. User Flow :
        i. The user opens the terminal simulator.
        ii. The user types a supported command into the input field.
        iii. The system validates the input and provides the corresponding response.
        iv. The result is then printed to the console and the user can type in more commands.
4. **Mini Desktop Simulator** - An interactive desktop simulation based on early graphical user interfaces of the 1980's. This shows how the user interacts with the desktop icons and the simple application windows, illustrating the evolution from command-line systems to visual desktop environments.
    a. Features :
        i. Draggable desktop icons - Users can move icons around the desktop.
        ii. Clickable applications - When you click on an icon, a window opens to show you information about it.
        iii. Window Management - Windows can be closed by a close button if they are opened.
        iv. Mouse interaction - Allows for drag-and-drop interaction using mouse events.
    b. User Flow :
        i. The user enters the desktop simulation and sees a desktop with icons.
        ii. The user clicks an icon to open its corresponding window.
        iii. Users drag icons on the desktop or move around open windows.
        iv. The user closes windows once they have finished looking through their contents.
5. **Voice Command** - An imitation voice assistant interface that mimics the current voice-controlled systems. Voice prompts are sent to the interface via a text box, and the interface responds with visual actions, reflecting the way in which modern digital assistants interpret and perform requests.
    a. Features :
        i. Voice command simulation - Users type into a text box and the text is converted to voice.
        ii. Interactive responses - Each response to a command has a different visual output and/or behavior in the interface.
        iii. Dynamic response area - Has a pulsing microphone icon and dynamic response area.
    b. User Flow :
        i. The user gives a voice command to the assistant interface.
        ii. The system checks the command and determines the appropriate response.

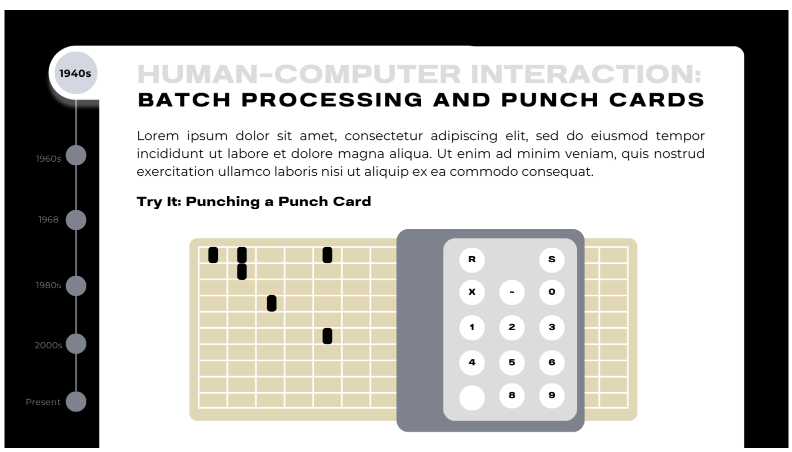
iii. The appropriate visual action is shown.
iv. Users keep experimenting with more commands and technologies.

IV. TENTATIVE STYLE GUIDE SNAPSHOT

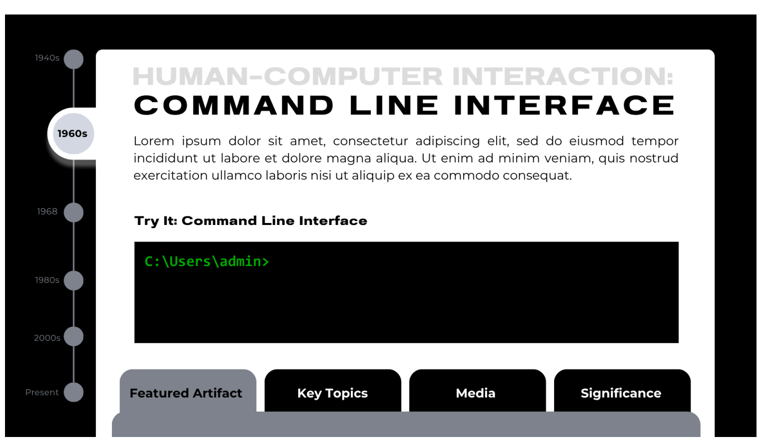
The images below are the layout and tentative design of the group's interactive element and exhibit.

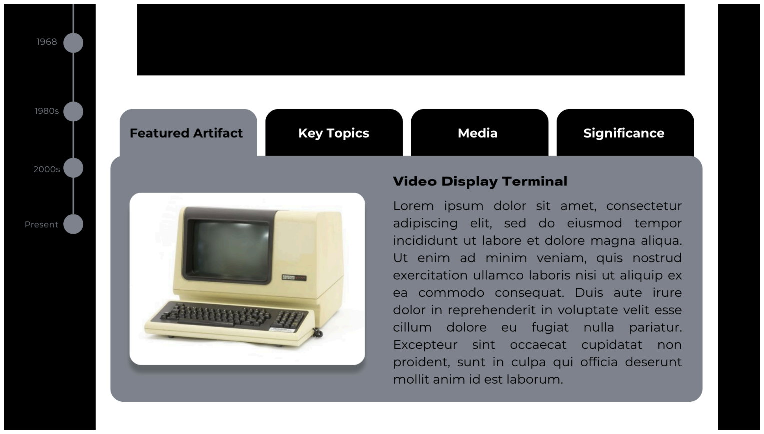


Layout 1: Milestone without simulative interactive elements. This layout applies to Pointing Devices, and Touch & Mobile Interfaces.

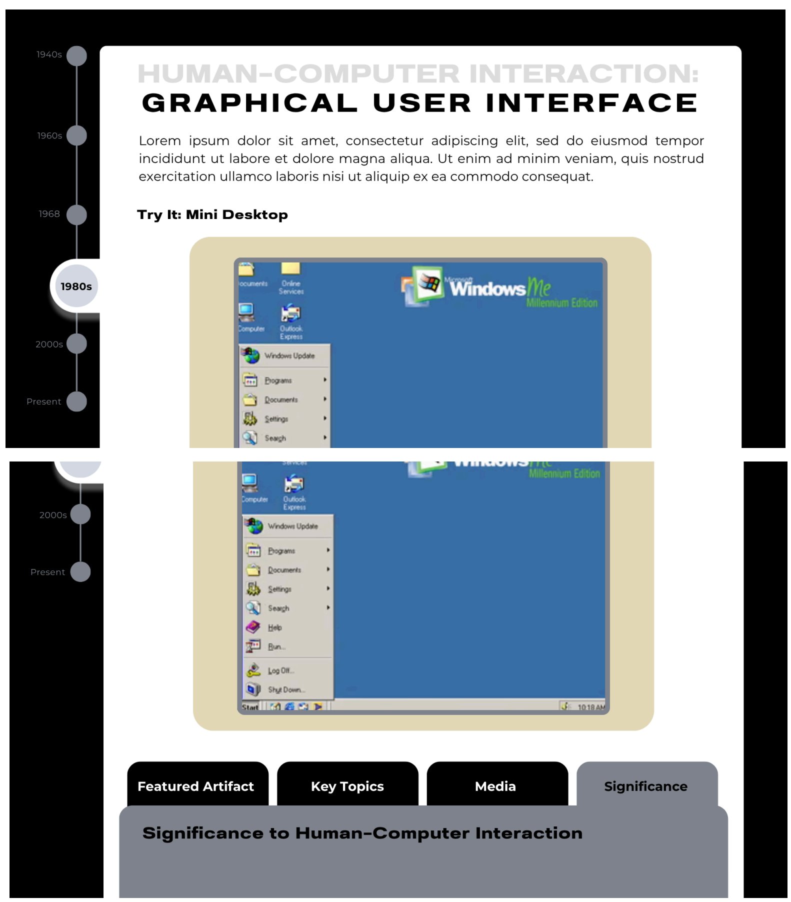


Layout 2: Batch Processing and Punch Cards' interactive element.

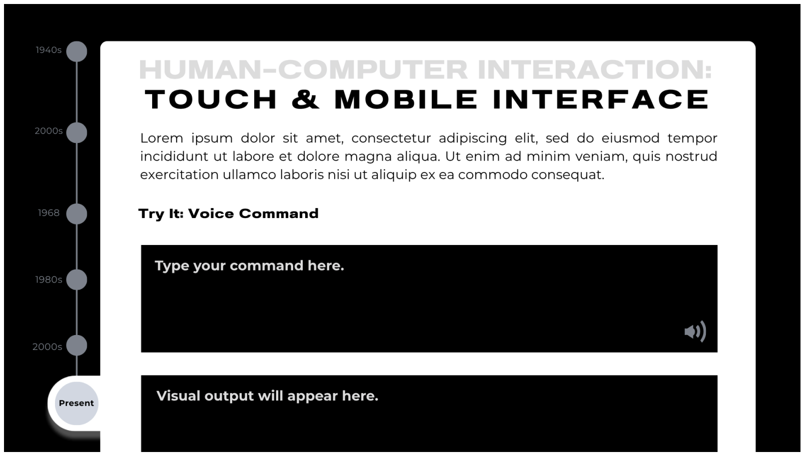


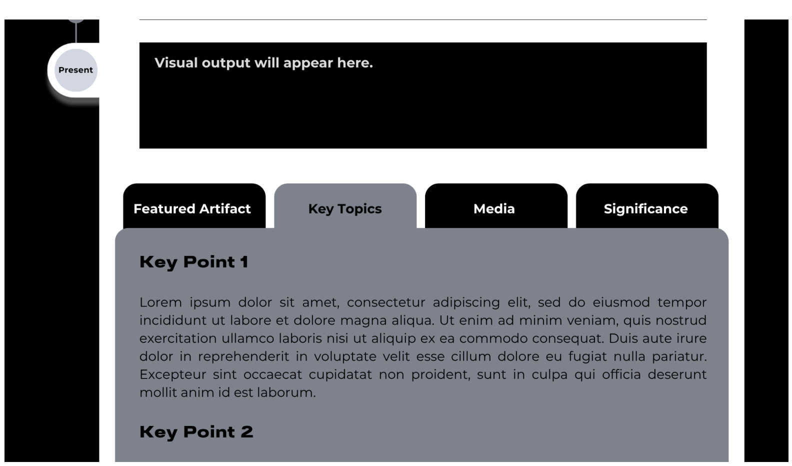


Layout 3: Command Line Interface interactive element and featured element layout.



Layout 4: Graphical User Interface's Mini Desktop interactive element and significance to HCI layout.





Layout 5: Touch & Mobile Interface' Voice Command interactive element and key topics layout.